APPENDIX

A BIG-O ANALYSIS OF THE ACCUMULATION ARRAY

Using just a single memory pool (i.e. with just an all array), the cost of insertion and sorting for each tensor component is

$$\sum_{i=1}^{N} O(\sum_{j=1}^{i} jlog(\sum_{j=1}^{i} j)) = \sum_{i=1}^{N} O(i^2 log(i^2)) = O(N^3 logN)$$

where N is the number of output nonzeros if the all array is sorted every time it is inserted, or

$$O(\sum_{j=1}^{N} jlog \sum_{j=1}^{N} j) = O(N^2 logN)$$

if the all array is sorted once. However, assuming the accumulation array is K times smaller than the all array, and the capacity does not change, the insertions into the all array are batched by the capacity of the accumulation array $\tilde{N} = N/K$, changing the cost to construct the output datastructure to

$$\sum_{k=1}^{K} \sum_{i=1}^{\tilde{N}} O(\sum_{j=1}^{i} jlog \sum_{j=1}^{i} j) = \sum_{k=1}^{K} \sum_{i=1}^{\tilde{N}} O(i^2 log(i^2)) = O(\tilde{N}^2 log\tilde{N}),$$

which is smaller than single memory pool by a constant factor.

B ADDITIONAL SPMTTKRP PERFORMANCE COMPARISON

Figure 25, Figure 26, and Figure 27 show SpMTTKRP runtime among different sorting algorithms on facebook, uber3, and nell-2 tensors, respectively.

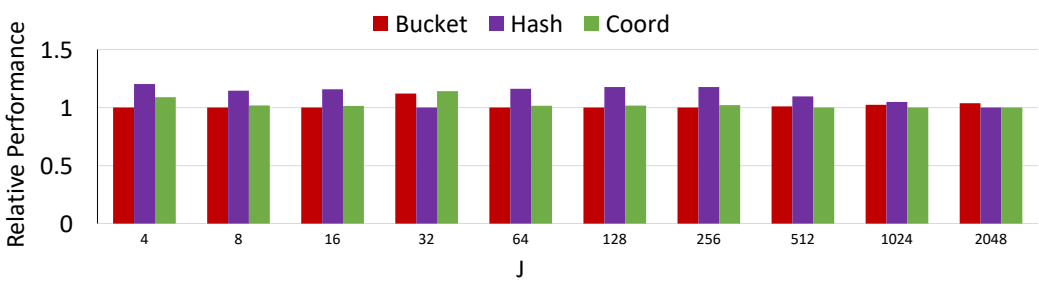

Fig. 25. facebook

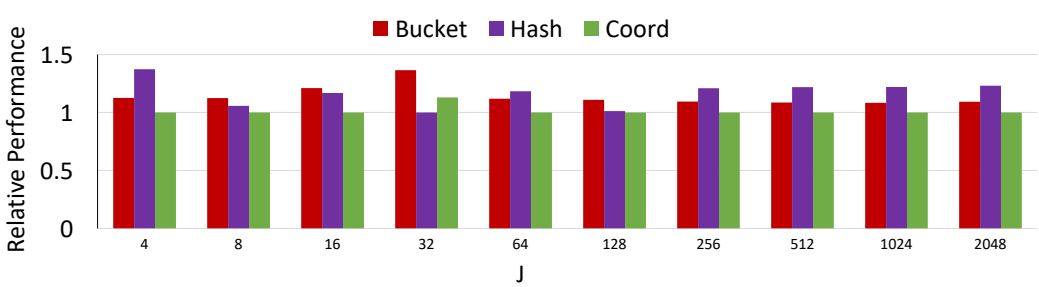

Fig. 26. uber3

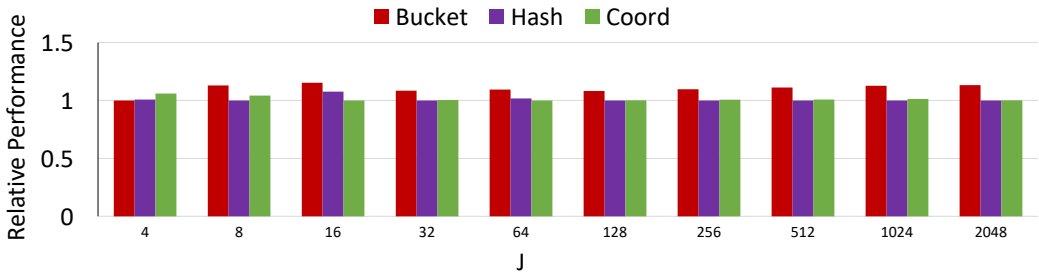

Fig. 27. Performance of various sparse workspace policies using nell-2

Table 4. Information on the selected tensors used to evaluate our work in Section 7.

Name	Dimension	NNZ	Sparsity	Dataset
uber3	183 × 24 × 1140	3309490	0.660988480	FROSTT [70]
nell-2	12092 × 9184 × 28818	76879419	0.000024022	FROSTT [70]
nips3	2482 × 2862 × 14036	3101609	0.000311080	FROSTT [70]
facebook	1504 × 42390 × 39986	737934	2.895E-7	Facebook [76]
freebase_sampled	38954435 × 38955429 × 532	139920771	1.733E-10	Freebase [33]

C TENSOR DATASET INFORMATION

Received 2023-11-16; accepted 2024-03-31

Table 5. Information on the selected matrices from SuiteSparse [42] used in Section 7.

Name	Dimension	NNZ	Sparsity	Kind
gams10am	114 × 171	407	0.020878219	linear programming problem
ch5-5-b4	120 × 600	600	0.008333333	combinatorial problem
lpi_cplex2	224 × 378	1215	0.01434949	linear programming problem
lp_scfxm1	330 × 600	2732	0.01379798	linear programming problem
odepb400	400 × 400	399	0.00249375	2D/3D problem
oscil_dcop_39	430 × 430	1544	0.00835046	subsequent circuit simulation problem
oscil_dcop_30	430 × 430	1544	0.00835046	subsequent circuit simulation problem
ex2	441 × 441	13640	0.070135386	computational fluid dynamics problem
lp_standmps	467 × 1274	3878	0.006518107	linear programming problem
west0497	497 × 497	1727	0.006991648	chemical process simulation problem
gre_512	512 × 512	2192	0.008361816	directed weighted graph
fs_760_3	760 × 760	5976	0.01034626	subsequent 2D/3D problem
bp_1400	822 × 822	4790	0.007089113	subsequent optimization problem
pde900	900 × 900	4380	0.005407407	2D/3D problem
n4c6-b14	920 × 6300	13800	0.002380952	combinatorial problem
lp_woodw	1098 × 8418	37487	0.004055734	linear programming problem
fpga_dcop_36	1220 × 1220	5892	0.003958613	subsequent circuit simulation problem
bcsstk12	1473 × 1473	17857	0.00823006	duplicate structural problem
freeFlyingRobot_3	1718 × 1718	6975	0.002363186	optimal control problem
adder_dcop_37	1813 × 1813	11246	0.003421389	subsequent circuit simulation problem
adder_dcop_21	1813 × 1813	11246	0.003421389	subsequent circuit simulation problem
CAG_mat1916	1916 × 1916	195985	0.053386546	combinatorial problem
G29	2000 × 2000	19990	0.009995	undirected weighted random graph
rail2586	2586 × 923269	8011362	0.003355441	linear programming problem
ex12	3973 × 3973	42092	0.005333256	computational fluid dynamics problem
ex12	3973 × 3973	42092	0.002666628	computational fluid dynamics problem
sts4098	4098 × 4098	38227	0.004552567	structural problem
M80PI_n	4182 × 4182	10261	0.000586707	eigenvalue/model reduction problem
t2dal_e	4257 × 4257	4257	0.000234907	duplicate model reduction problem
seymourl	4944 × 6316	38493	0.001232711	linear programming problem
n4c6-b13	6300 × 25605	88200	0.000546768	combinatorial problem
c-36	7479 × 7479	36710	0.000656292	optimization problem
bcsstm38	8032 × 8032	7842	0.000121557	structural problem
nemeth16	9506 × 9506	298259	0.003300638	subsequent theoretical/quantum chemistry problem
flowmeter5	9669 × 9669	67391	0.00072084	model reduction problem
sit100	10262 × 10262	34094	0.000323753	2D/3D problem
c-42	10471 × 10471	60378	0.000550684	optimization problem
c-44	10728 × 10728	47864	0.000831767	optimization problem
ch7-6-b3	12600 × 4200	50400	0.000952381	combinatorial problem
bayer10	13436 × 13436	94926	0.00052583	chemical process simulation problem
cyl6	13681 × 13681	363961	0.0038891	structural problem
ch7-6-b4	15120 × 12600	75600	0.000396825	combinatorial problem
psse0	26722 × 11028	102432	0.000347592	power network problem
2D_27628_bjtcai	27628 × 27628	442898	0.00058023	semiconductor device problem
c-55	32780 × 32780	218115	0.000405973	optimization problem
scagr7-2r	32847 × 46679	120141	0.000078356	linear programming problem sequence
stat96v3	33841 × 1113780	3317736	0.000088023	linear programming problem
jan99jac100sc	34454 × 34454	215862	0.000181843	economic problem
mosfet2	46994 × 46994	1499460	0.000678969	semiconductor device problem
Andrews	60000 × 60000	410077	0.00011391	computer graphics/vision problem
GaAsH6	61349 × 61349	1721579	0.000457417	theoretical/quantum chemistry problem
lhr71	70304 × 70304	1528092	0.000309164	chemical process simulation problem
oilpan	73752 × 73752	1835470	0.000337442	structural problem
t3dh_a	79171 × 79171	2215638	0.000353481	duplicate model reduction problem
x104	108384 × 108384	5138004	0.000437385	structural problem
cage12	130228 × 130228	2032536	0.000119848	directed weighted graph
tp-6	142752 × 1014301	11537419	0.000079681	linear programming problem
marine1	400320 × 400320	6226538	0.000038853	chemical oceanography problem
t2em	921632 × 921632	4590832	0.000005405	electromagnetics problem
G3_circuit	1585478 × 1585478	4623152	0.000001839	circuit simulation problem